RoME: Robust Mixture of Low-Rank Experts against Multiple Adversarial Perturbations

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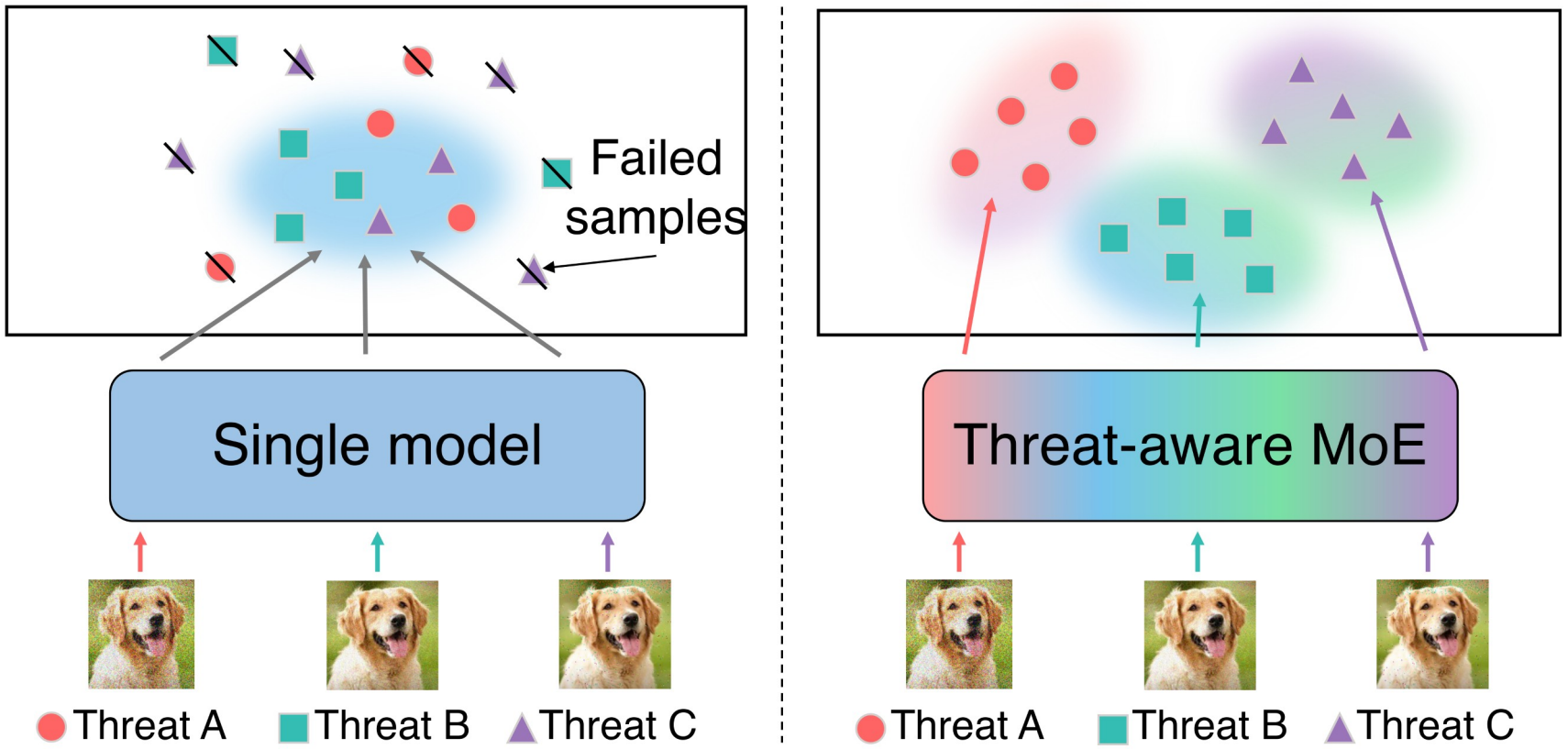


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github.com/wkim97/RoME



Multi-Perturbation Robustness

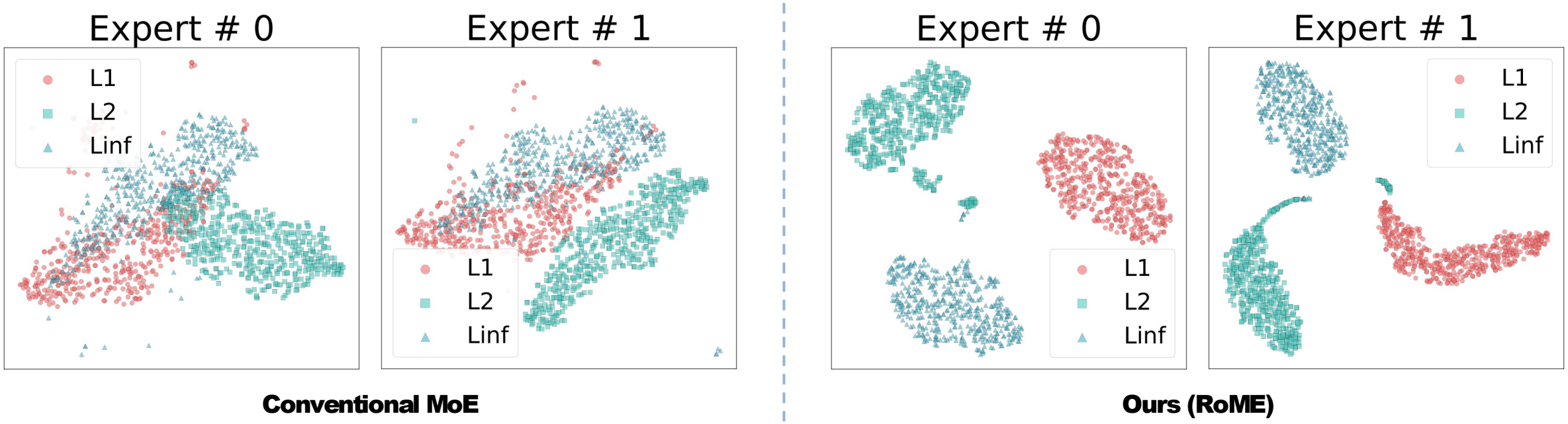


- AT on one threat leaves the model **vulnerable to different threats**
- AT on multiple threats (MAT) causes **cross-threat trade-off**
- Our idea: give each threat **its own model pathway** via a **mixture of experts (MoE)**

Why Naïve MoE Fails

① Redundant Experts

- Experts redundantly capture **threat-common features**

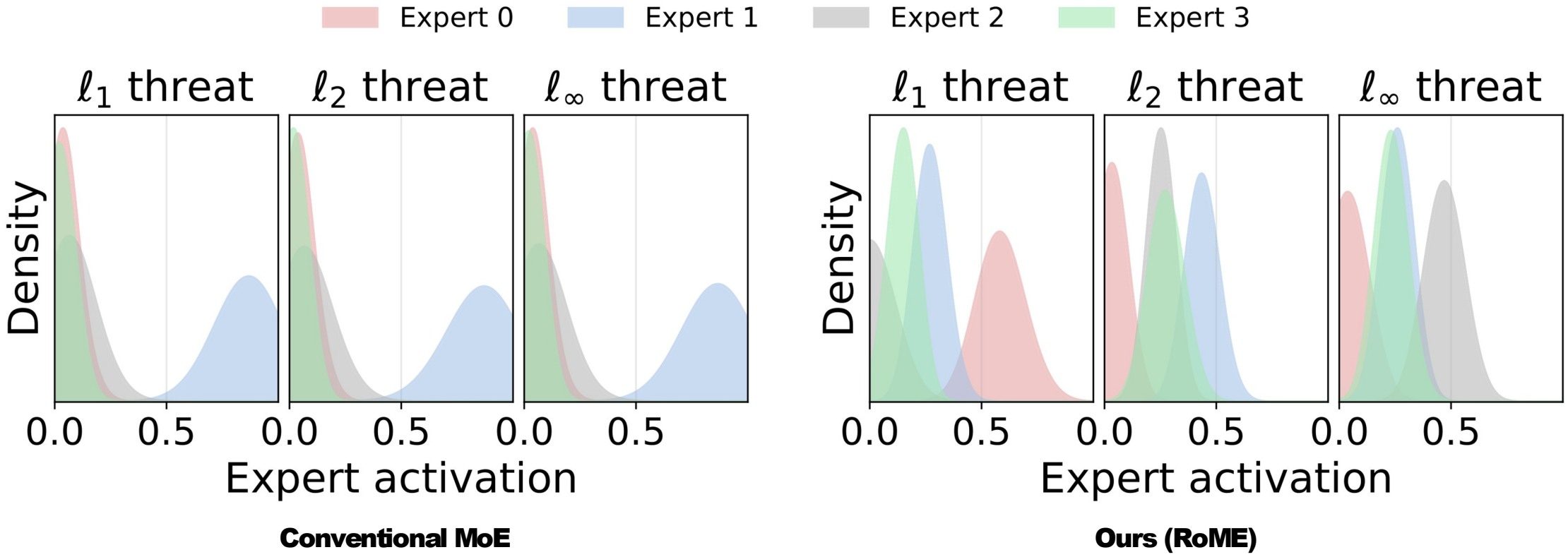


Conventional MoE **Ours (RoME)**

Why Naïve MoE Fails

② Threat-Agnostic Routing

- Gating routes each threat to **nearly identical experts**



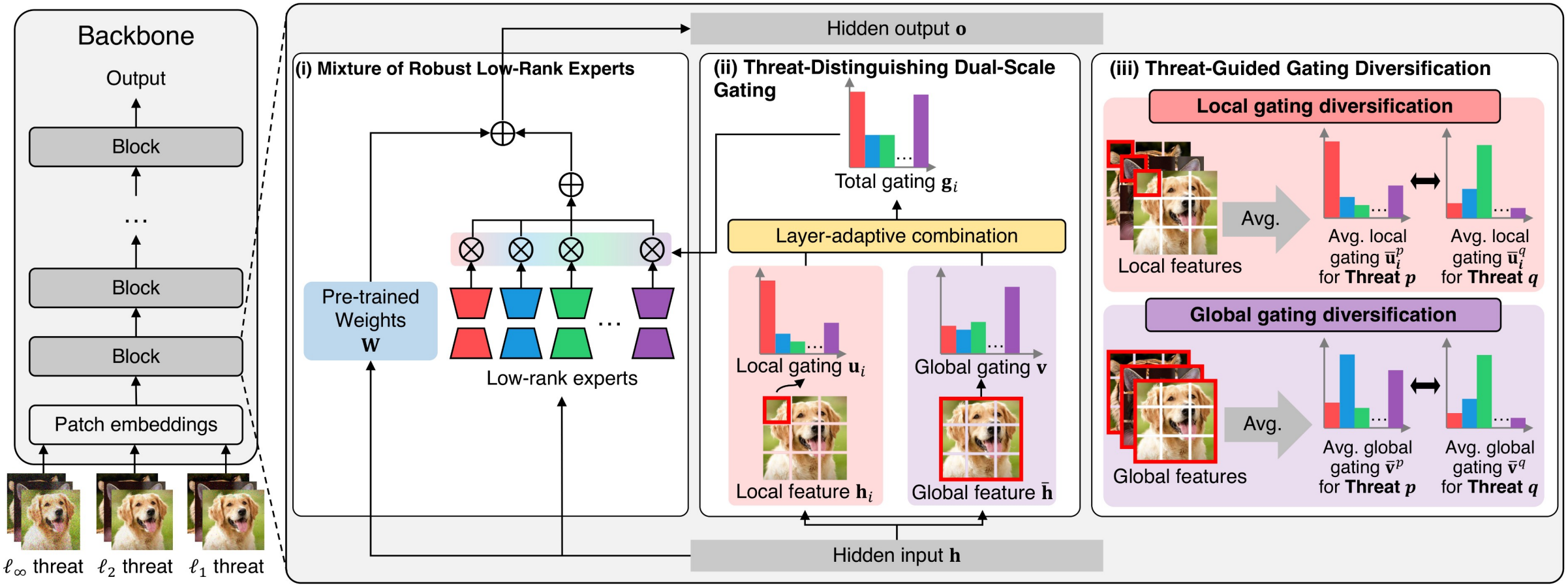
Conventional MoE **Ours (RoME)**

→ Lacks **specialization of each expert** on different threats

RoME: Mixture of Low-Rank Experts

Overall Framework

Low-rank experts on a shared backbone + **dual-scale gating** + **gating diversification**



RoME: Mixture of Low-Rank Experts

(i) Mixture of Low-Rank Experts

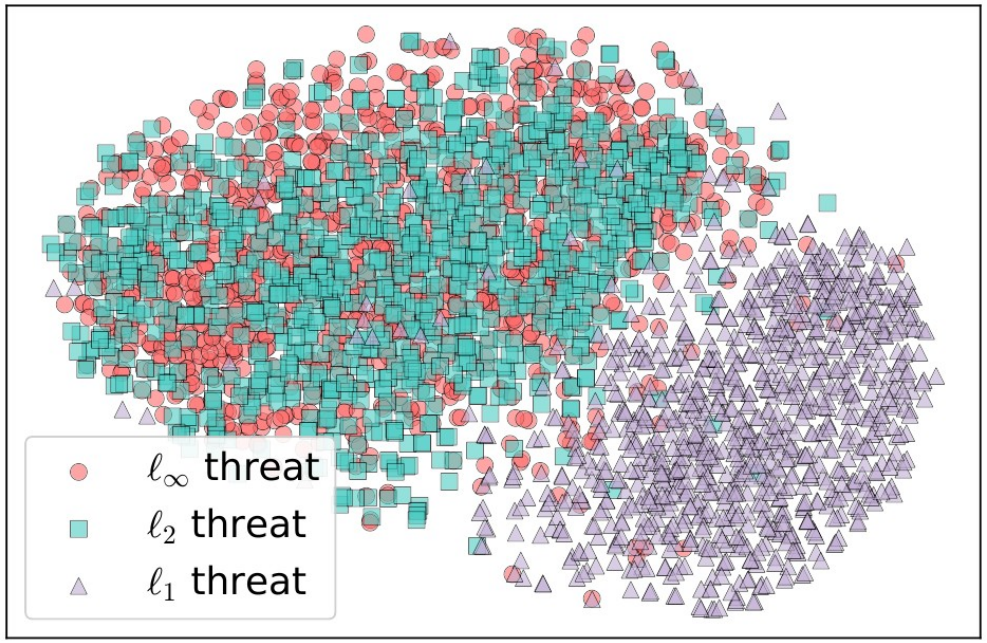


Conventional MoE **Ours (RoME)**

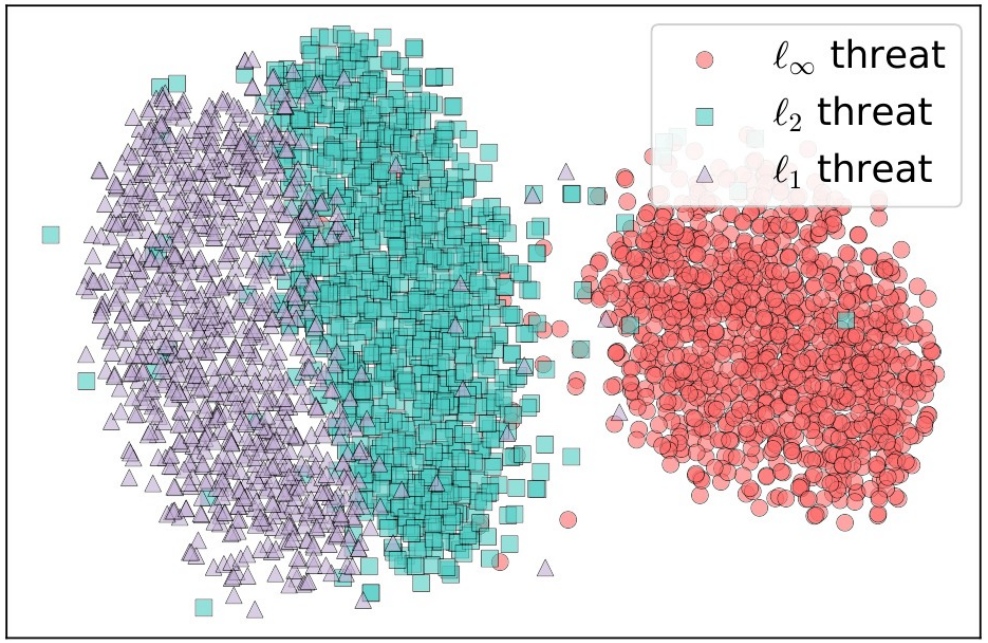
- **Low-rank (LoRA) experts** on the shared backbone
- Backbone: **threat-common** features
- Low-rank experts: **threat-specific** features
- → ① **Threat-specialized experts**

RoME: Mixture of Low-Rank Experts

(ii) Threat-Distinguishing Dual-Scale Gating



(a) Patch-level

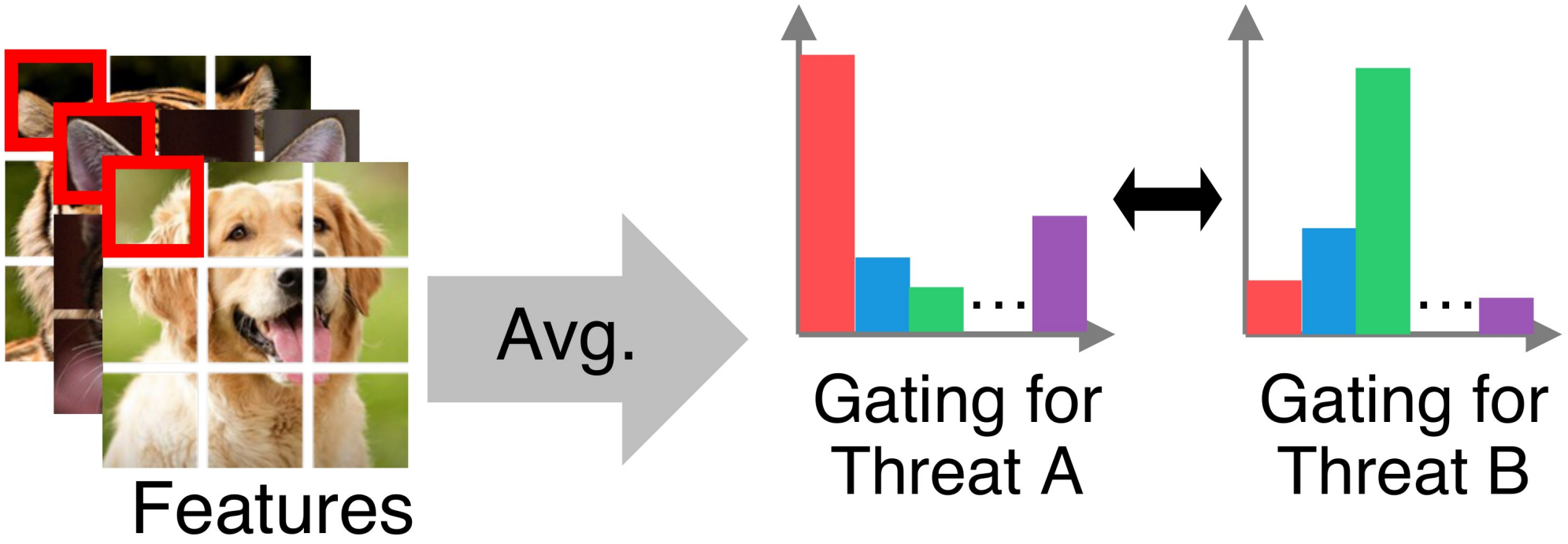


(b) Image-level

- Each threat is distinguishable at **different levels of granularity**
- **Dual-scale gating** at local (patch) and global (image) level
- → ② **Threat-aware routing**

RoME: Mixture of Low-Rank Experts

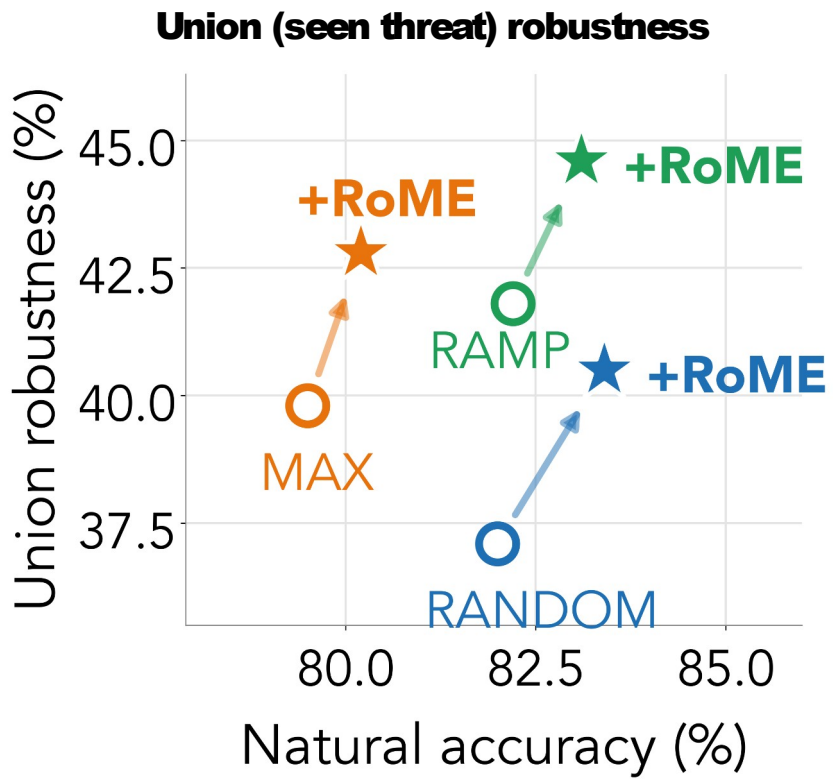
(iii) Threat-Guided Gating Diversification

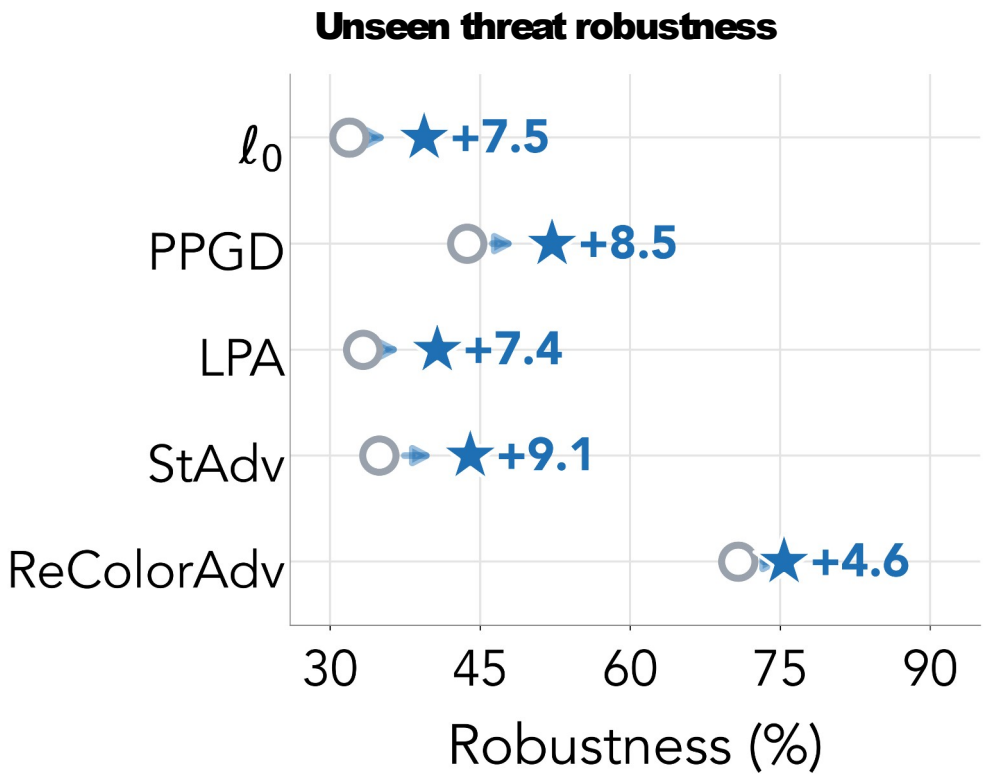


- Maximise **pairwise distance** between per-threat gating patterns
- Enforces **distinct expert combinations** for each threat
- → ② **Threat-aware routing**

Experimental Evaluations

Robustness

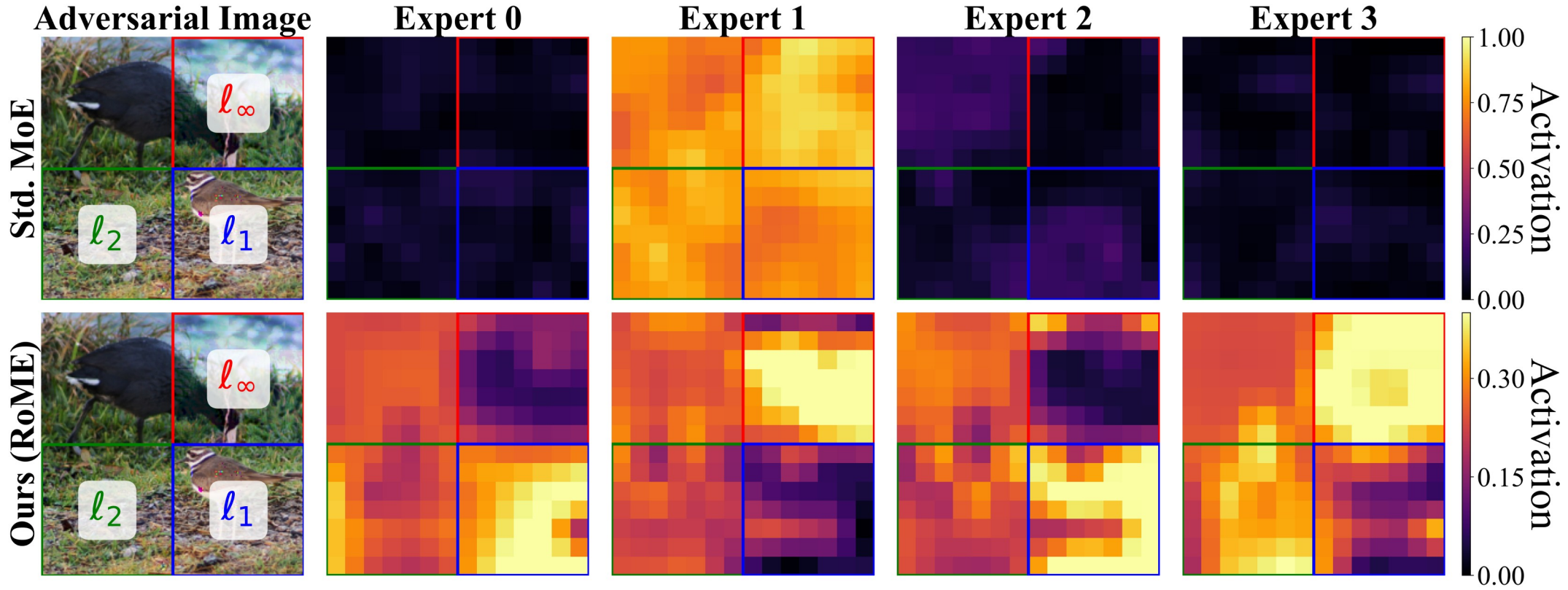




- Improves **union robustness and natural accuracy** on different MAT methods
- Also improves robustness against **threats unseen during training**

Experimental Evaluations

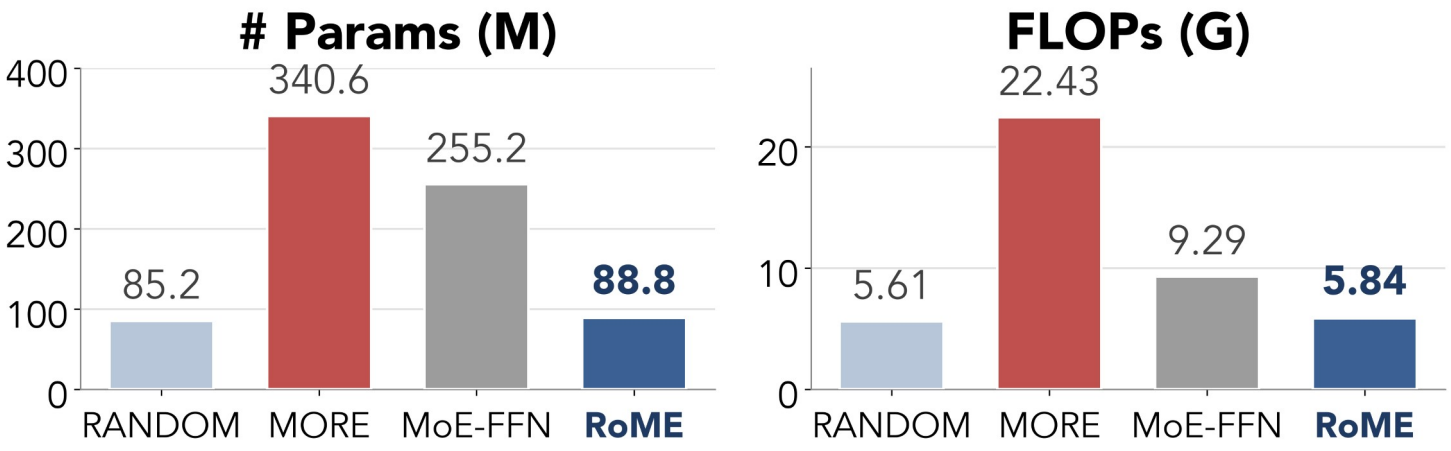
Learned Expert Routing

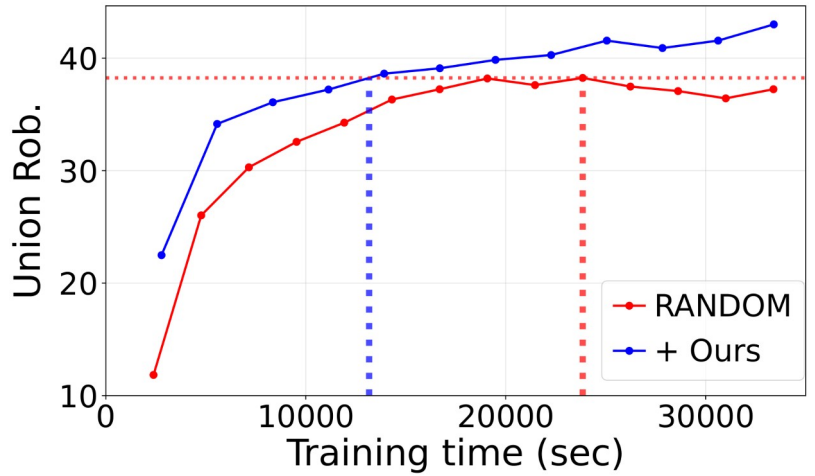


- Standard MoE activates **the same experts** for each threat
- RoME activates **different experts per threat**

Experimental Evaluations

Computational Cost





- SoTA union robustness at only **1.04× params** and **1.04× FLOPs**
- Reaches the baseline's peak robustness in **~1.8× less training time**

Summary

- Naïve MoE fails on multiple threats: **redundant experts** and **threat-agnostic routing**
- **RoME** = **low-rank experts** on a shared backbone + **dual-scale gating** + **gating diversification**
- **SoTA union robustness and natural accuracy**, plus better **unseen-threat robustness**
- Only **1.04× params and 1.04× FLOPs** over the MAT baseline

github.com/wkim97/RoME

Thank you!